

Multiple Choice (10 marks)

1. Which of the following statements about glucose metabolism is correct?
 - (a) The end-product of glycolysis in all cells is glucose
 - (b) Glycolysis doesn't require any enzymes
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) Glucose metabolism only occurs in the presence of oxygen
 - (e) In glycolysis, glucose is converted into two molecules of fructose
2. Which of the following vitamins is required for fatty acid b-oxidation?
 - (a) Vitamin C
 - (b) Vitamin E
 - (c) Vitamin B 12
 - (d) Thiamin
 - (e) Riboflavin
3. An example of a discontinuous change over the adult years is the change in
 - (a) Personality
 - (b) Skin elasticity
 - (c) Physical strength
 - (d) Male hormone levels
 - (e) Hair color
4. Deficiency of vitamin A in children causes:
 - (a) Poor muscle development
 - (b) Increased risk of mortality
 - (c) Goitre
 - (d) Increased risk of obesity
 - (e) Increased risk of heart disease
5. Disease can most often be prevented by which of the following:
 - (a) Sunscreen
 - (b) Vaccines
 - (c) Antivirals
 - (d) Pain Relievers
 - (e) Hand Sanitizer

True / False (10 marks)

Answer True or False

6. True or False: Familial hypercholesterolaemia is caused by mutations in the gene that encodes high density lipoprotein.
7. True or False: Phosphocreatine is predominantly found in the mitochondria of muscle cells, facilitating ATP production during aerobic respiration.
8. True or False: Placing a stoma directly on the umbilicus is recommended for improved patient comfort and care.
9. True or False: Astroviruses cause widespread replication in the liver and can lead to liver infections.
10. True or False: Hepatitis C was not first identified through clinical observation, but rather through serological testing methods.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the various limitations of food balance sheets in assessing nutritional health and food security, providing a critical analysis of their impact on public health policy.
12. Discuss the effectiveness of various treatments for depression in older adults, specifically analyzing why mental stimulation may not be considered an effective intervention.

End of Paper